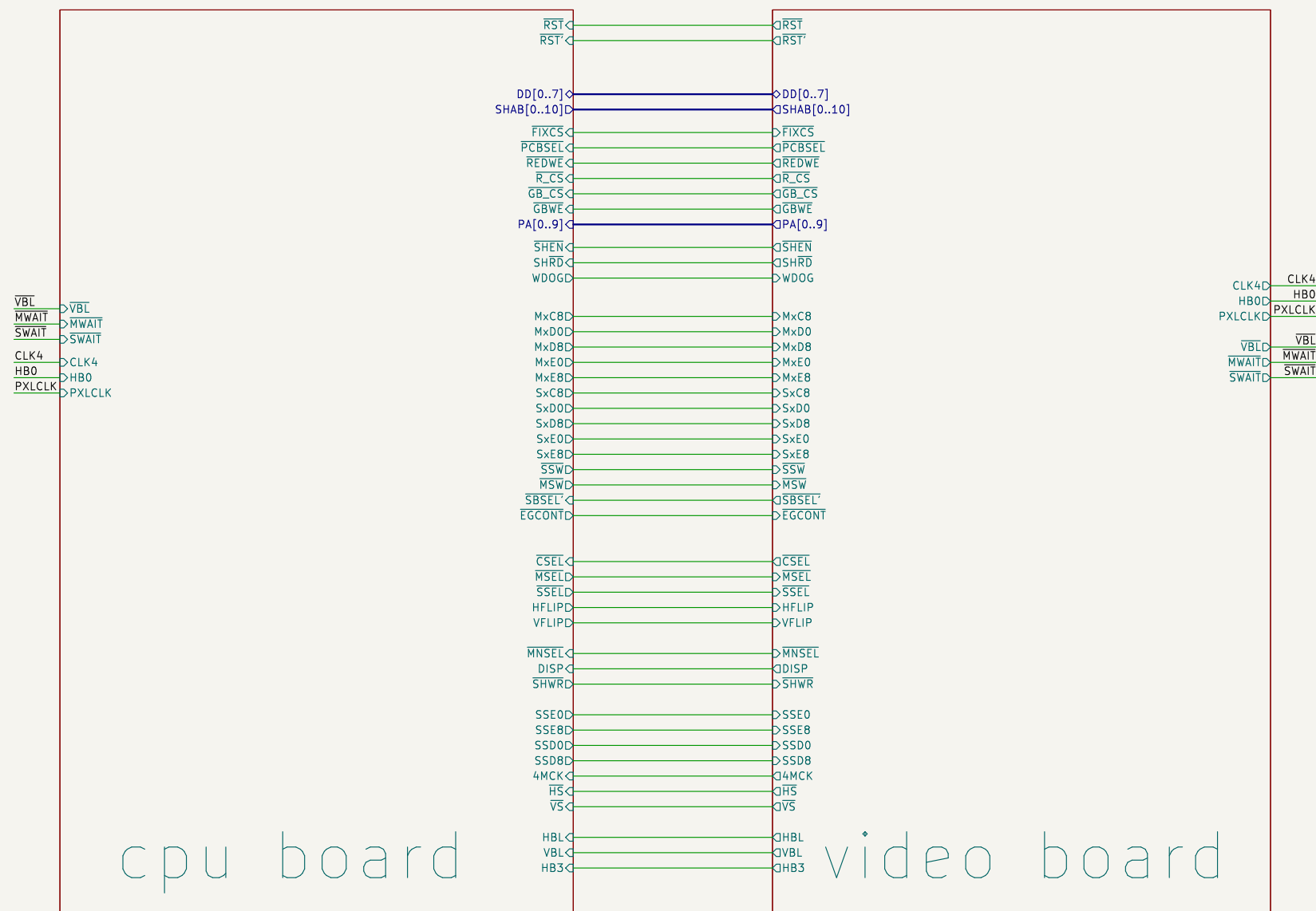




cpu board

video board

```
tehkanwch
bootleg board
```

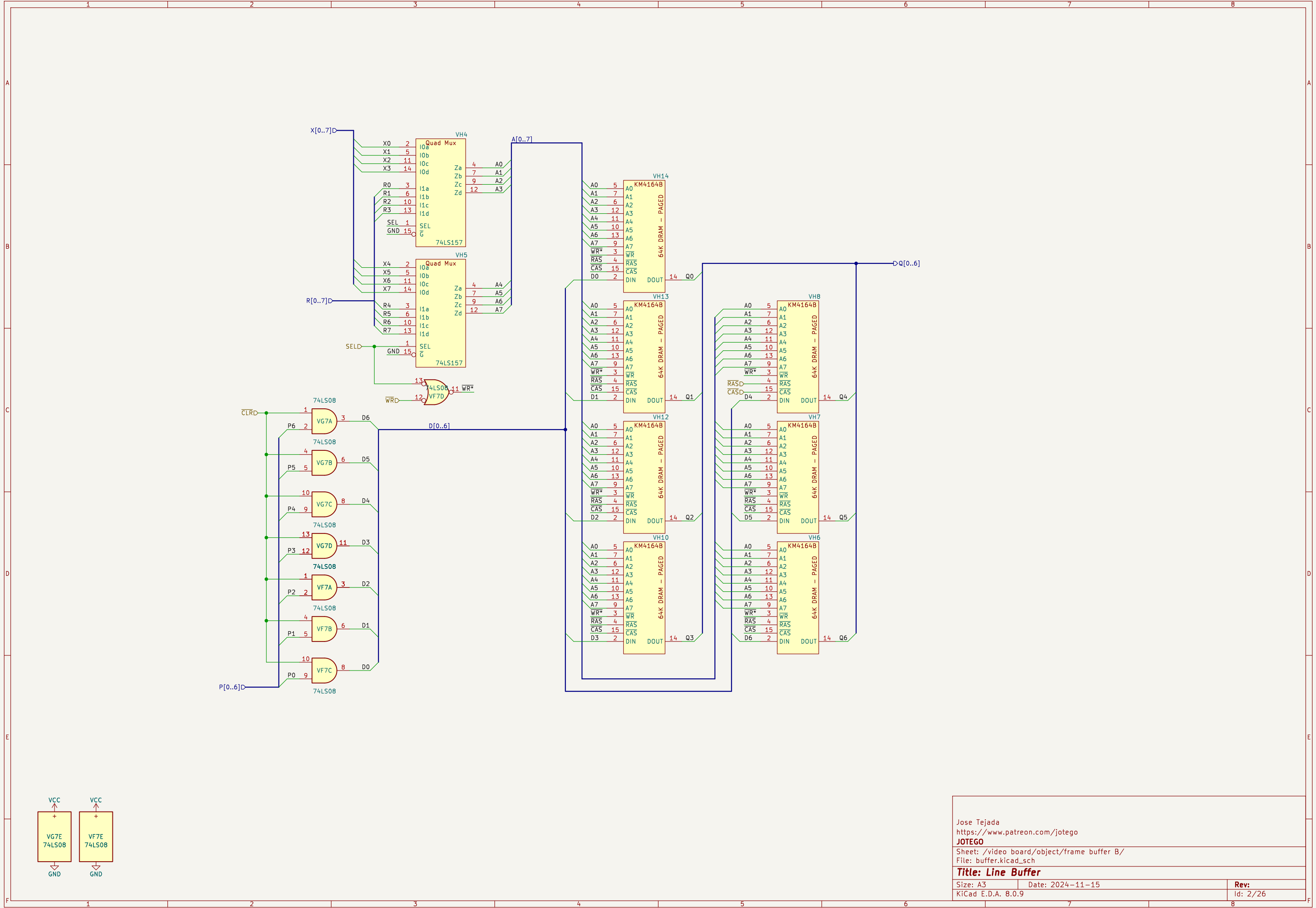
- original PCB has through-hole vias whereas the bootleg has filled vias
 - original uses three custom packages from Mitsubishi with logic inside.
- These were replaced by a small daughter board on the bootleg, serving the same functionality
- different edge connectors
 - debug edge connector on video board not present on bootleg
 - cabinet input logic partly missing and rewired on the bootleg to replace the trackball interface for a joystick one
 - different power amplifier for sound without heat sink on the bootleg
 - slightly different frequencies for crystal oscillators (cost reduction?)
 - device reference markers (silkscreen) on bootleg have reversed order on bootleg

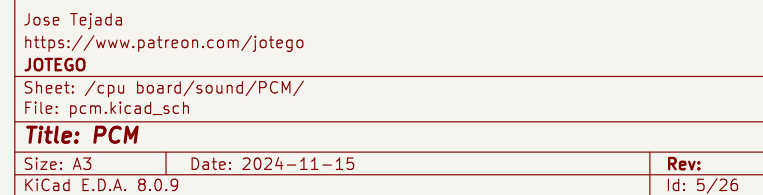
Jose Tejada
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 Sheet: /
 File: tehkanwch.kicad_sch

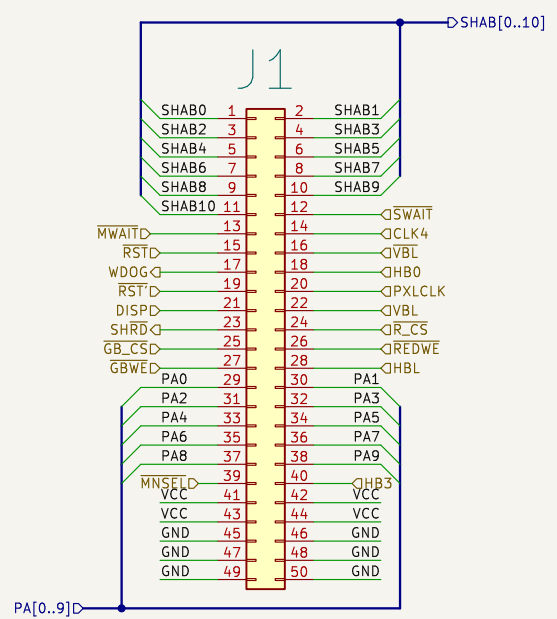
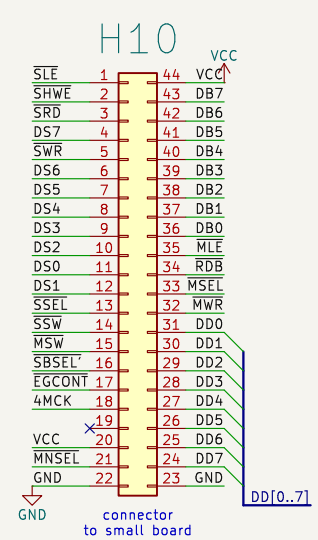


Title: TEHKAN WORLD CUP (tehkanwch bootleg)

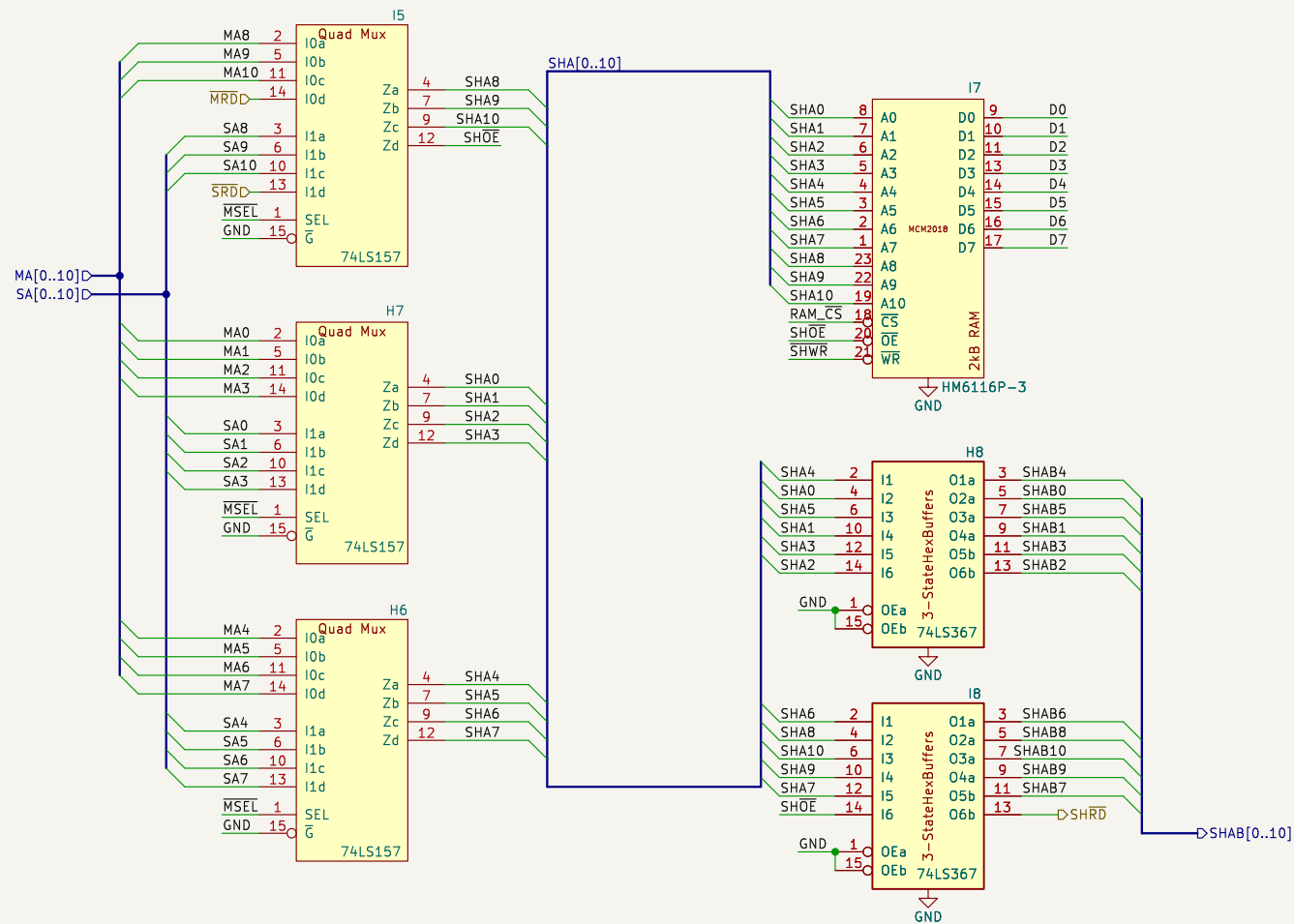
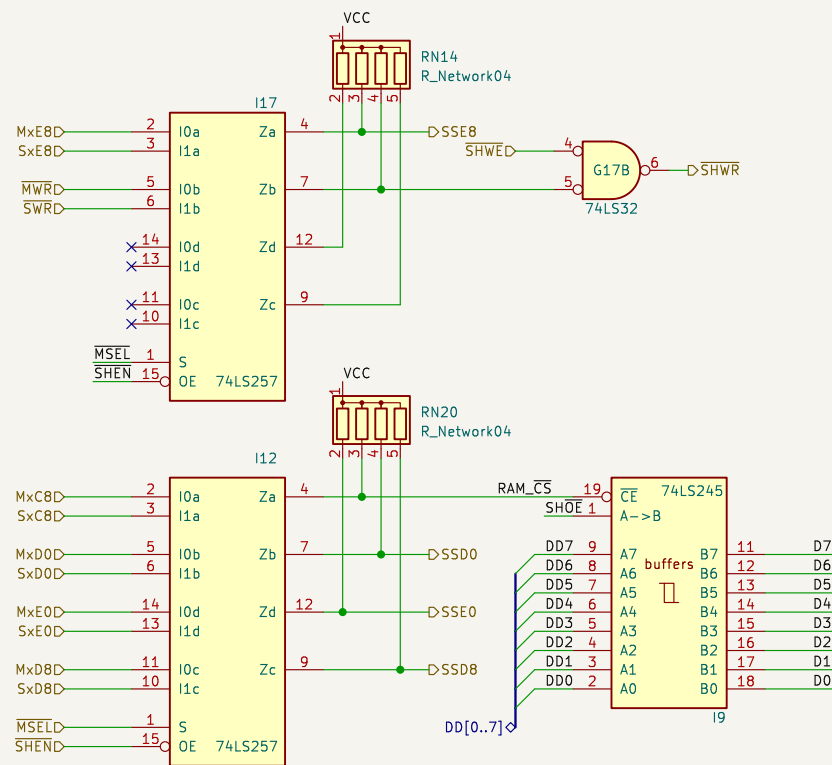
Size: A3	Date: 2024-11-15	Rev:
KiCad E.D.A. 8.0.9		Id: 1/26

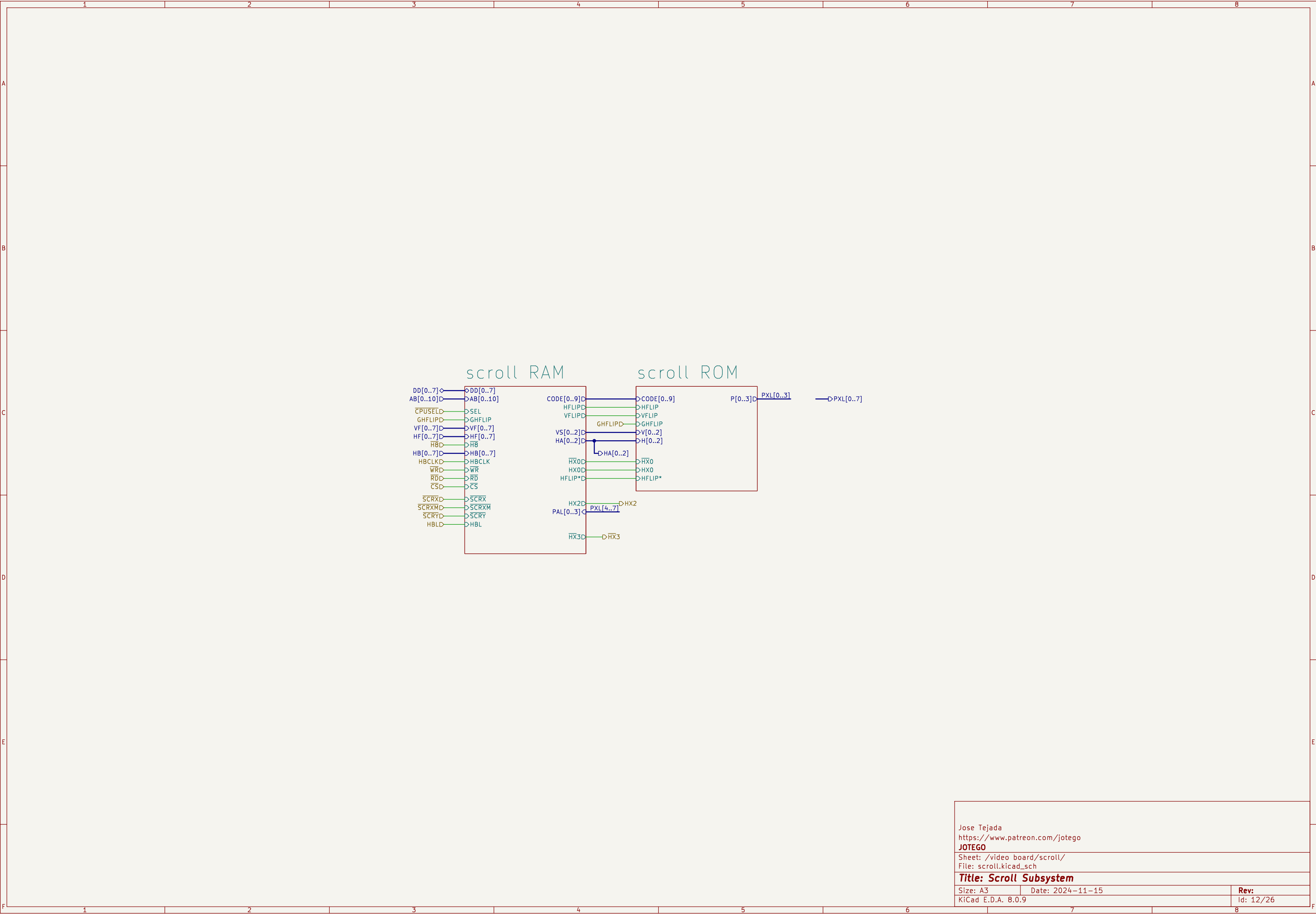


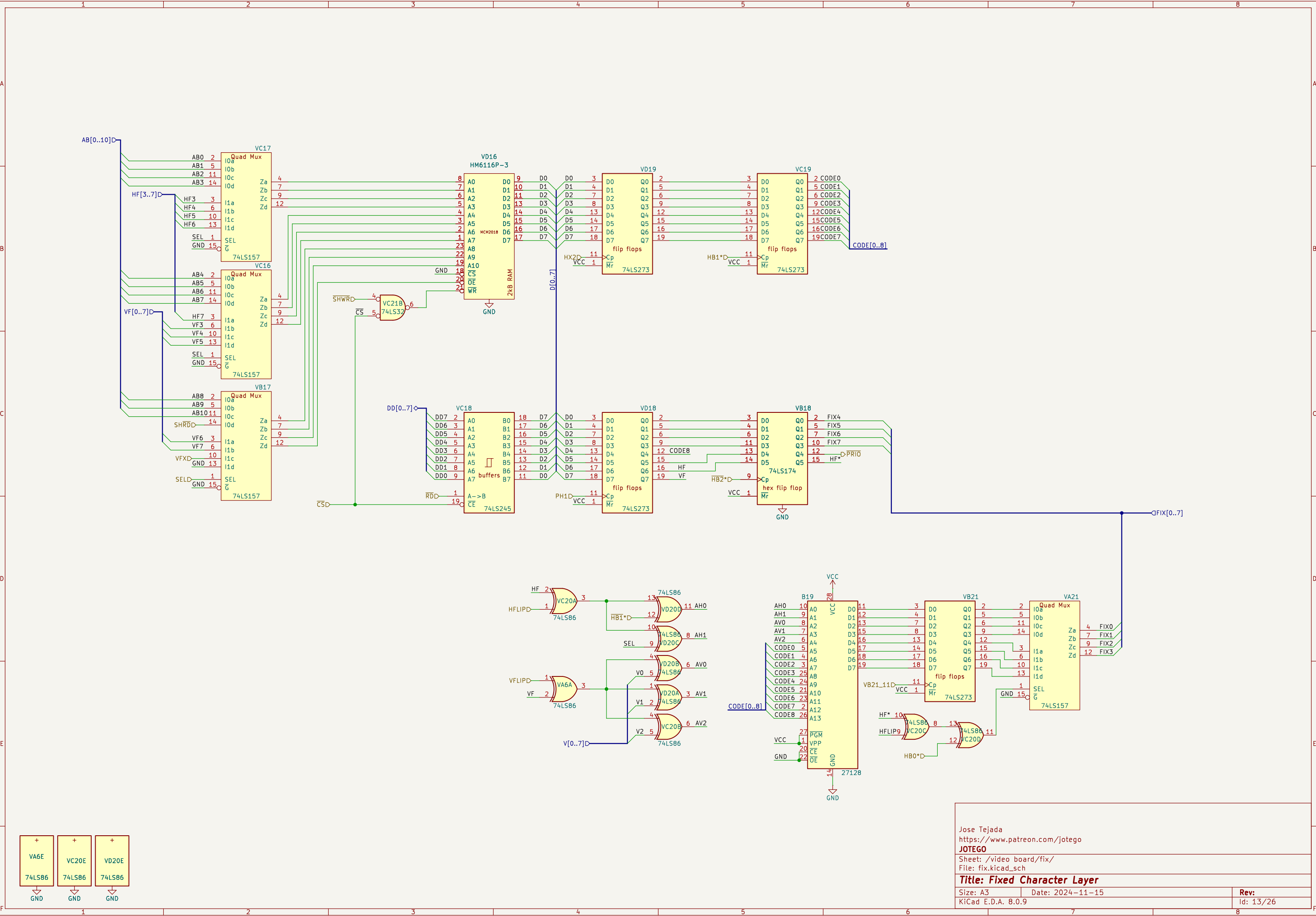




Timing diagram for daughter.kicad_sch. The diagram shows the relationship between various signals and registers. The signals are: 4MCK, SRD, SWR, RDB, MWR, MLE, SLE, SBSEL', and MNSSEL. The registers are: DD[0..7], DB[0..7], DS[0..7], SHWE, SSW, MSWD, EGCONT, MSEL, SSEL, and EGCONTC. The diagram shows that 4MCK is the clock for all registers. SRD, SWR, RDB, MWR, MLE, and SLE are data inputs to the registers. SBSEL' and MNSSEL are control signals for the registers. The diagram shows that the registers are updated on the rising edge of 4MCK.







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Sheet: /video board/fix/
File: fix.kicad_sch

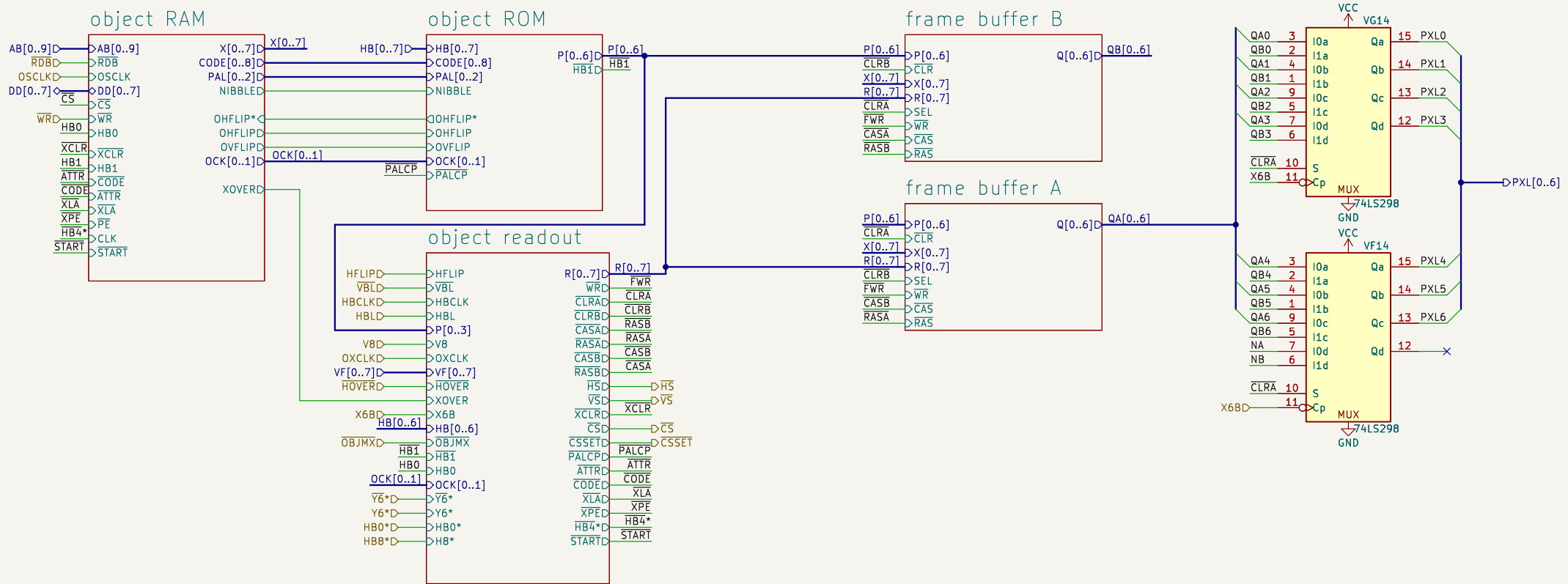
Title: Fixed Character Layer

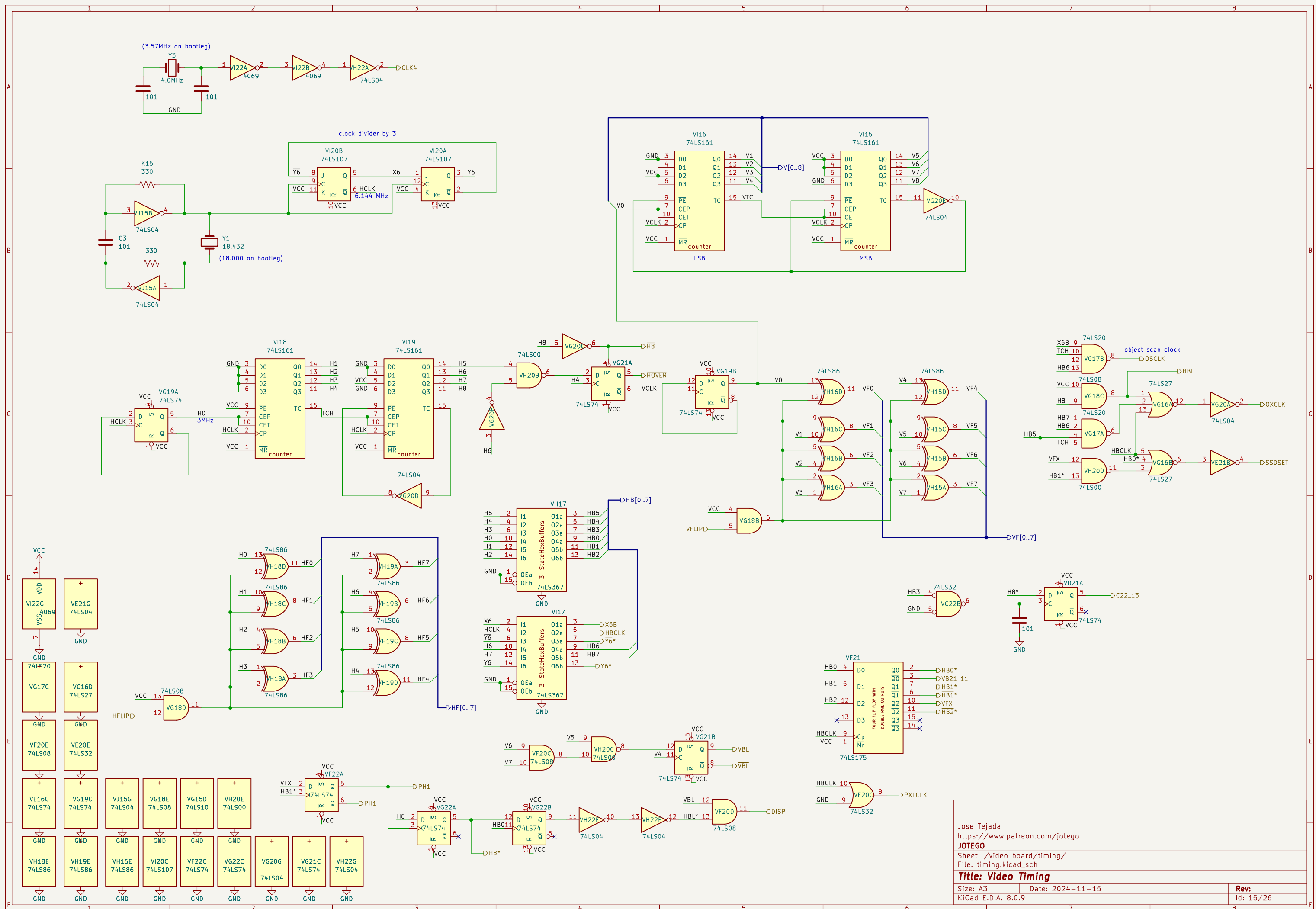
Size: A3 Date: 2024-11-15

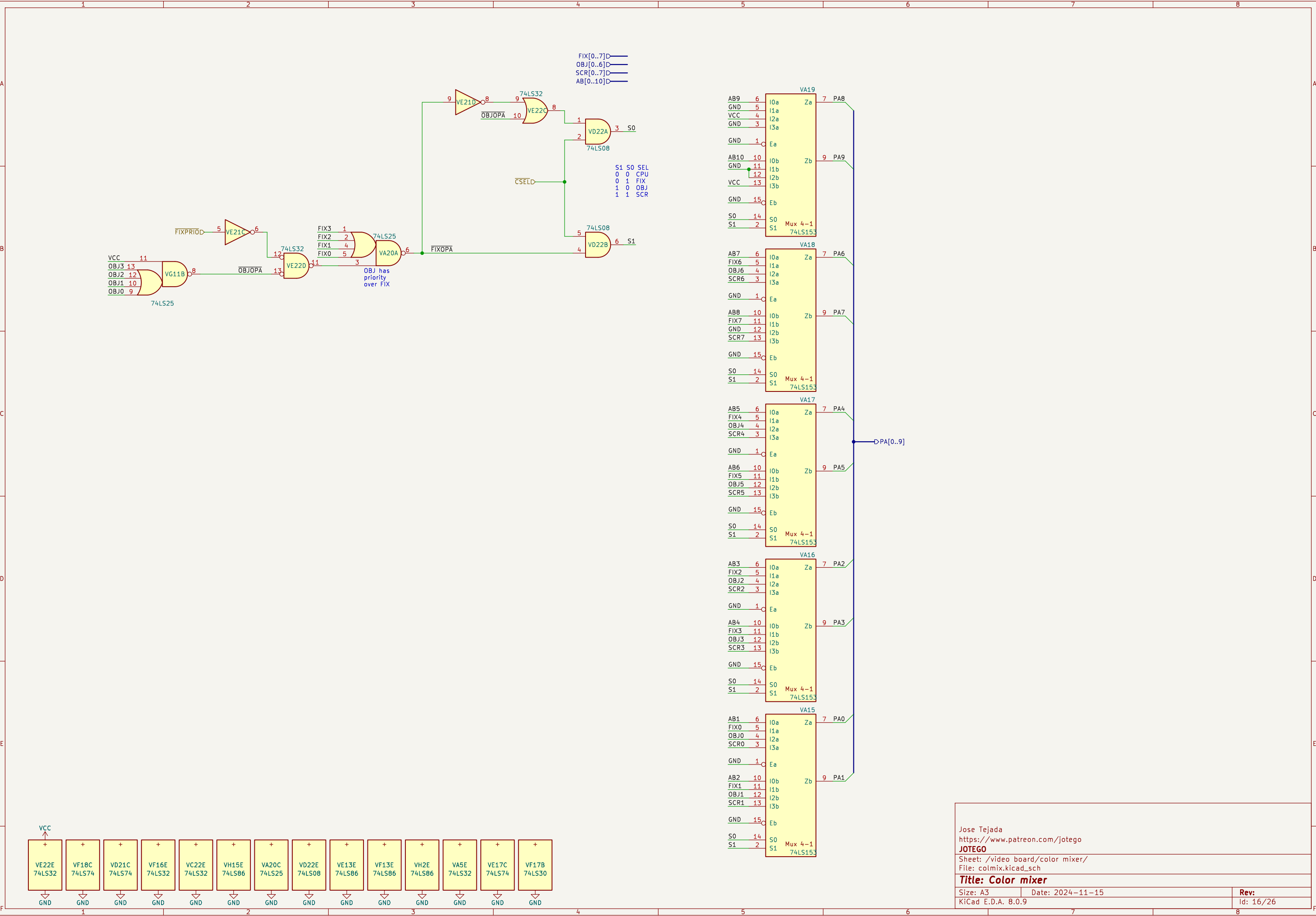
Rev:

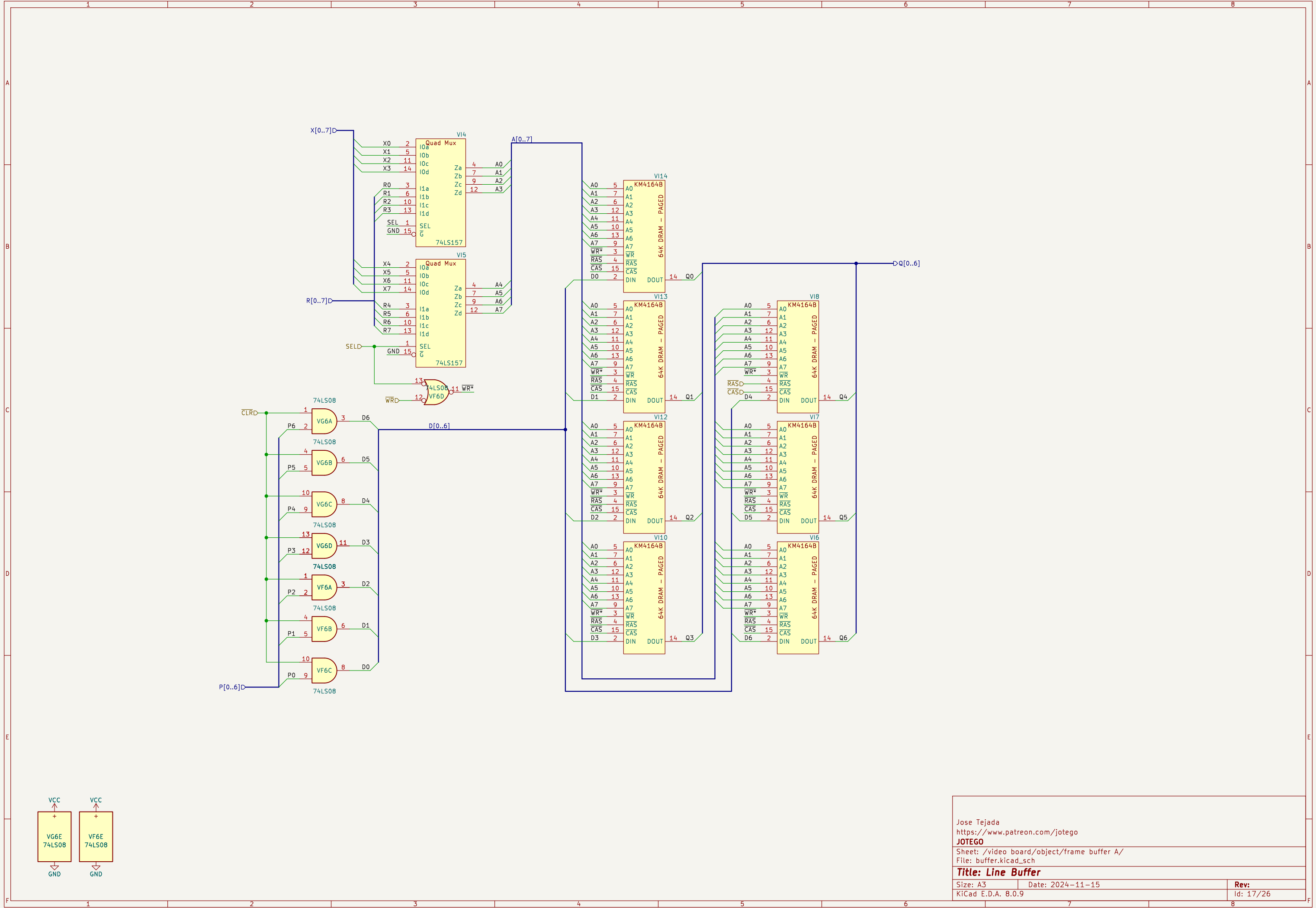
KiCad E.D.A. 8.0.9

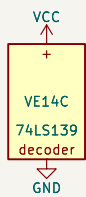
Id: 13/26



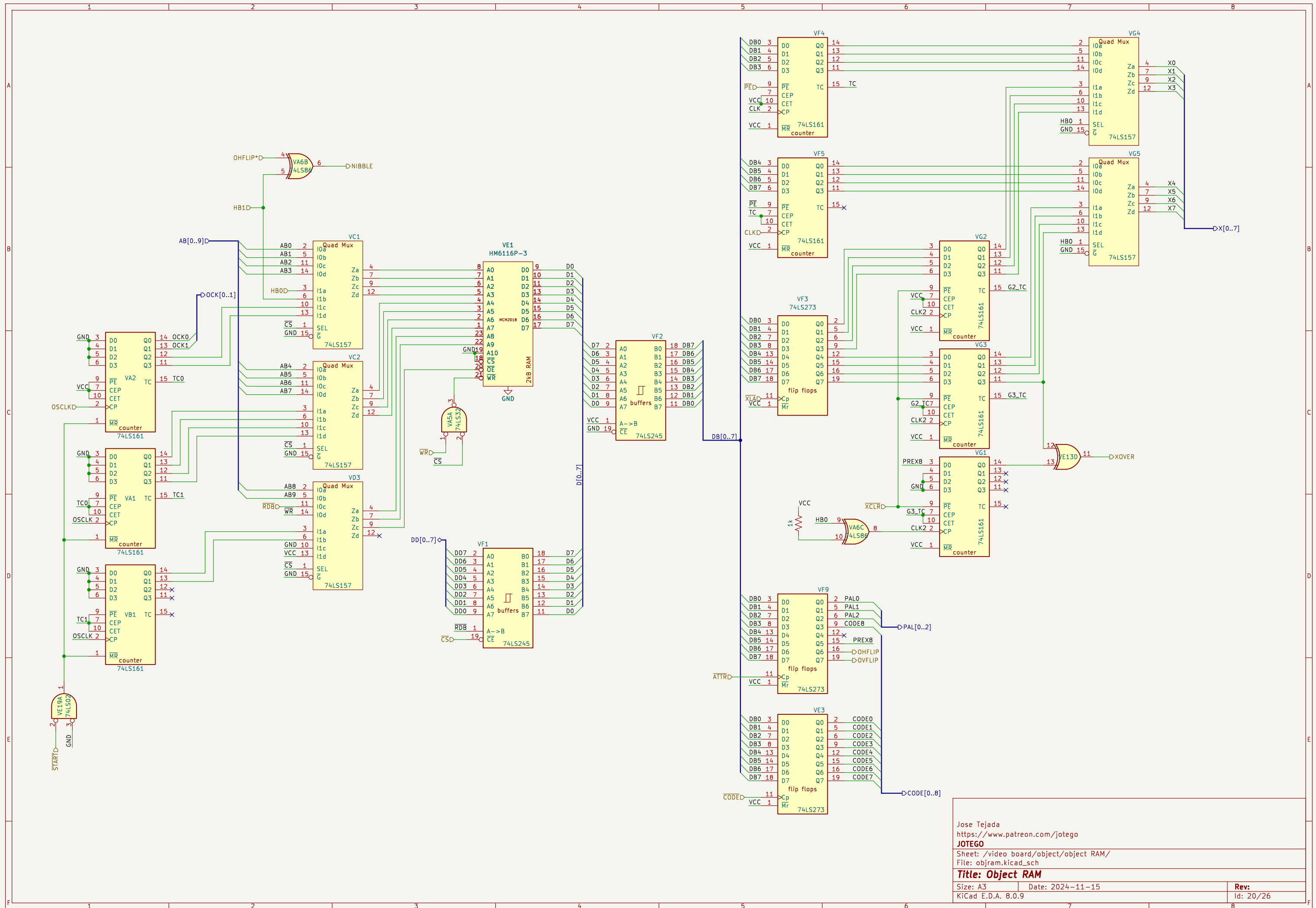








Rev:
Id: 19/26



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Sheet: /video board/object/object RAM/
File: objram.kicad_sch

Title: Object RAM

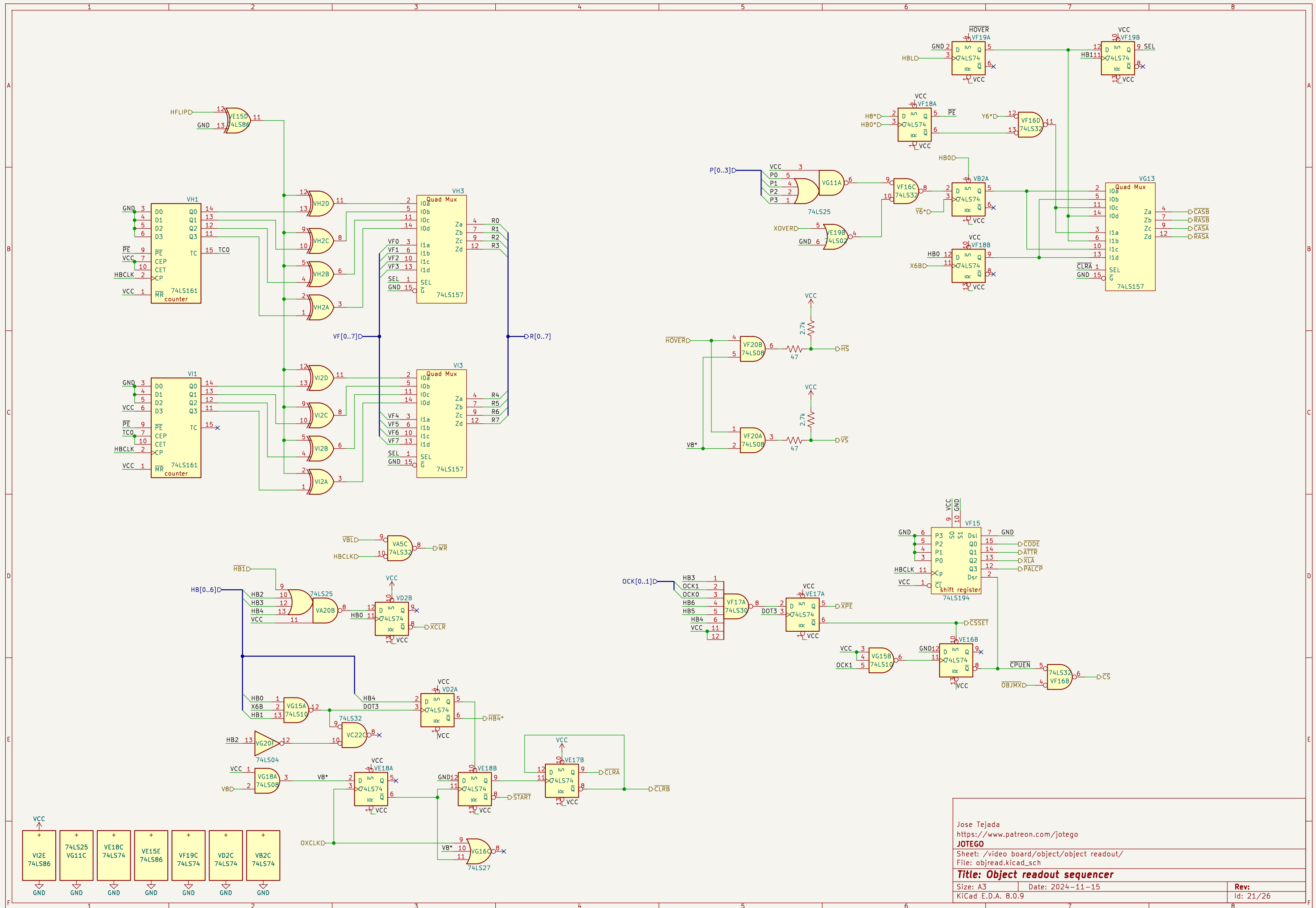
Size: A3

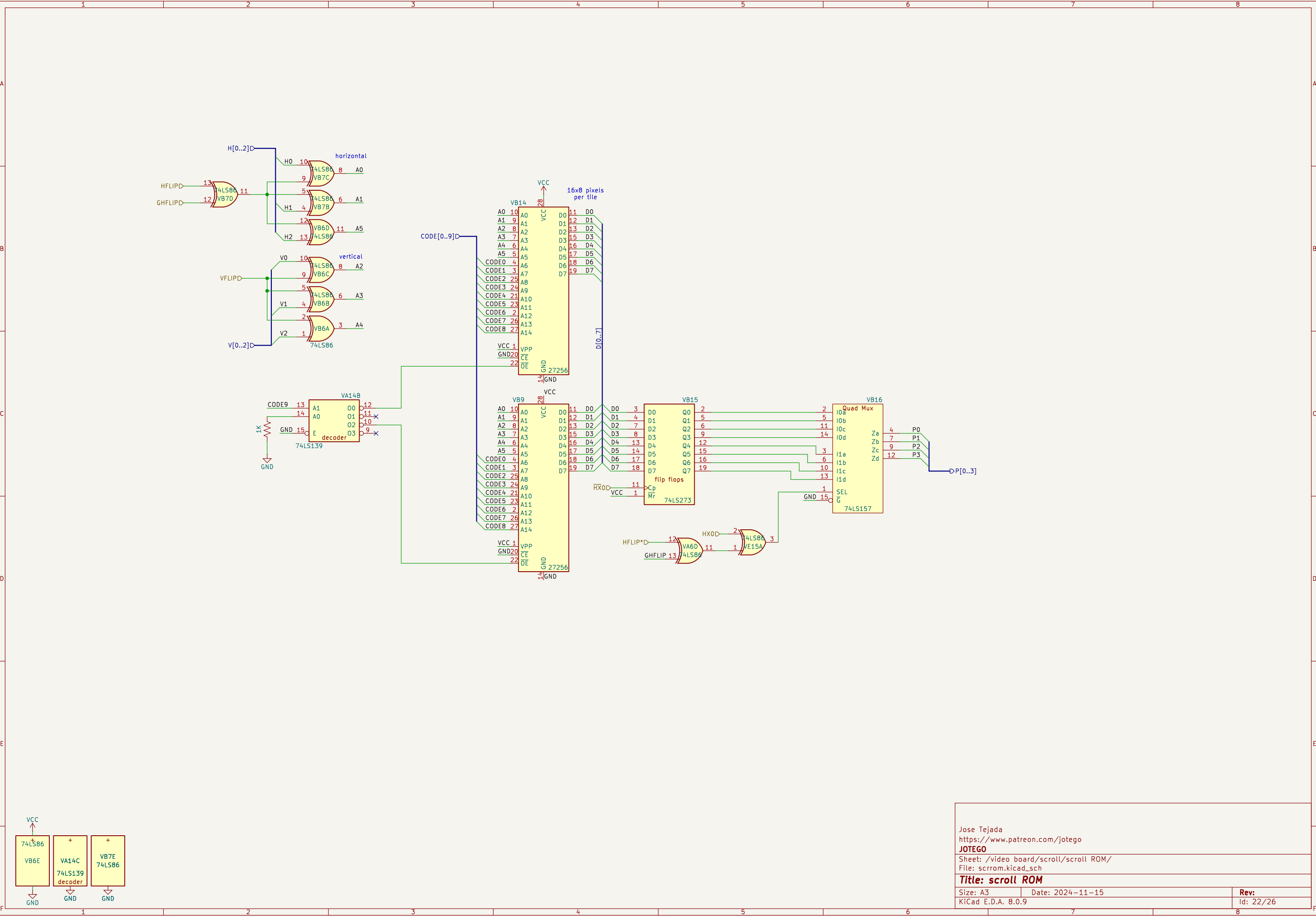
Date: 2024-11-15

Rev:

KiCad E.D.A. 8.0.9

Id: 20/26





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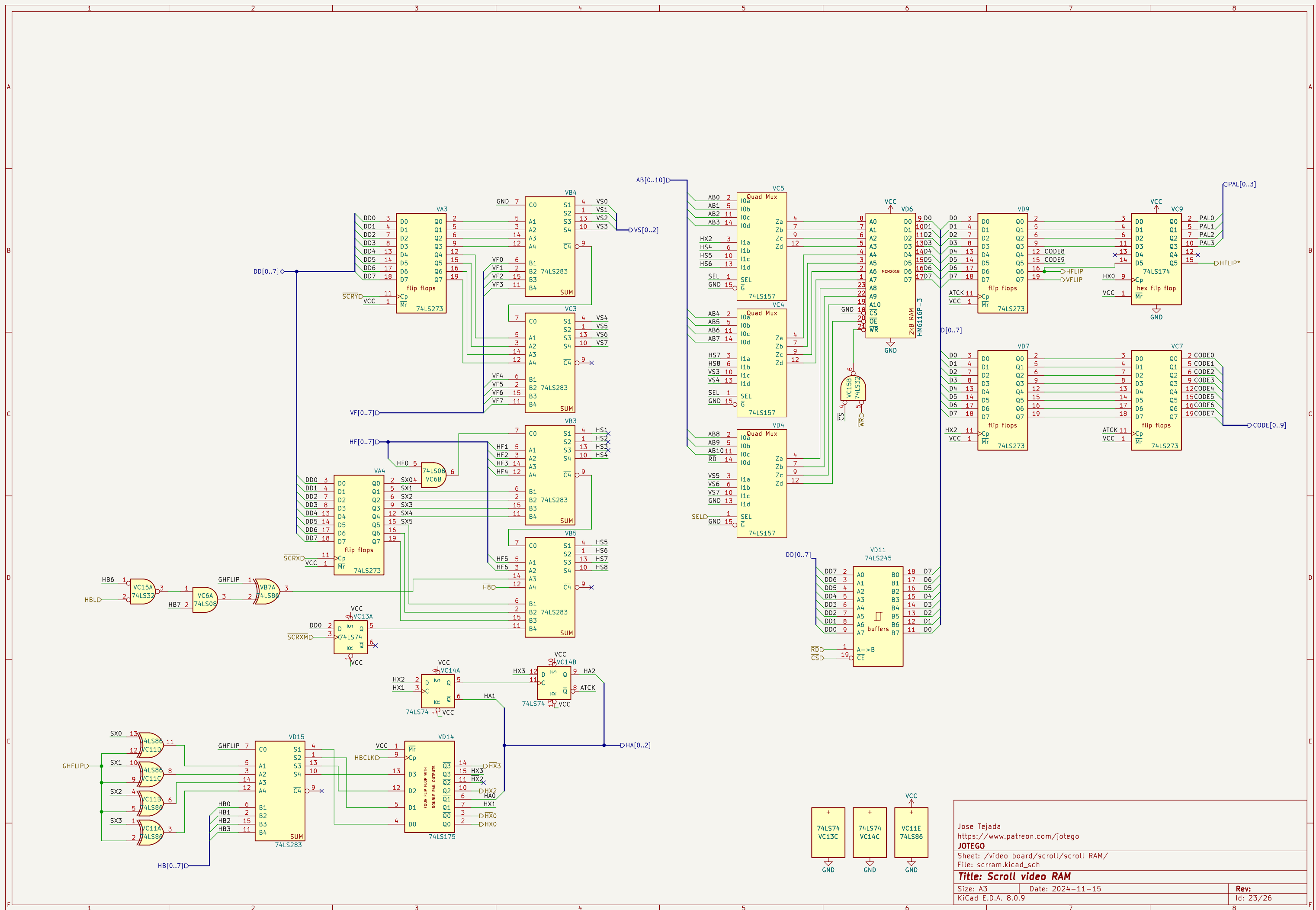
Sheet: /video board/scroll ROM/
File: scrrom.kicad_sch

Title: scroll ROM

Size: A3 Date: 2024-11-15

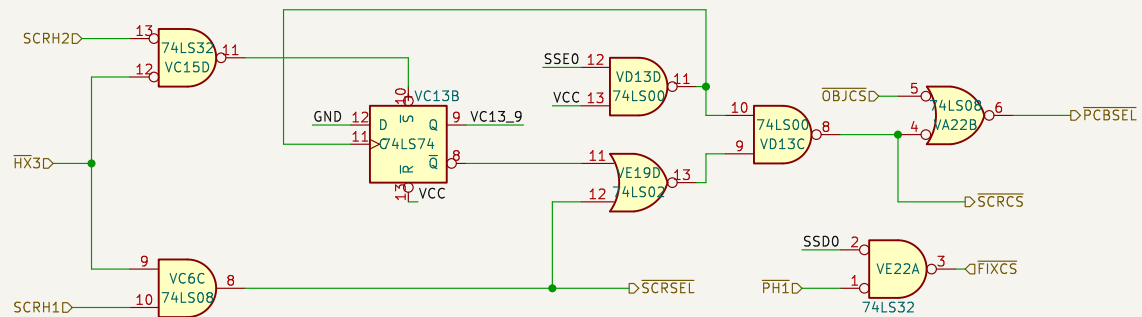
KiCad E.D.A. 8.0.9

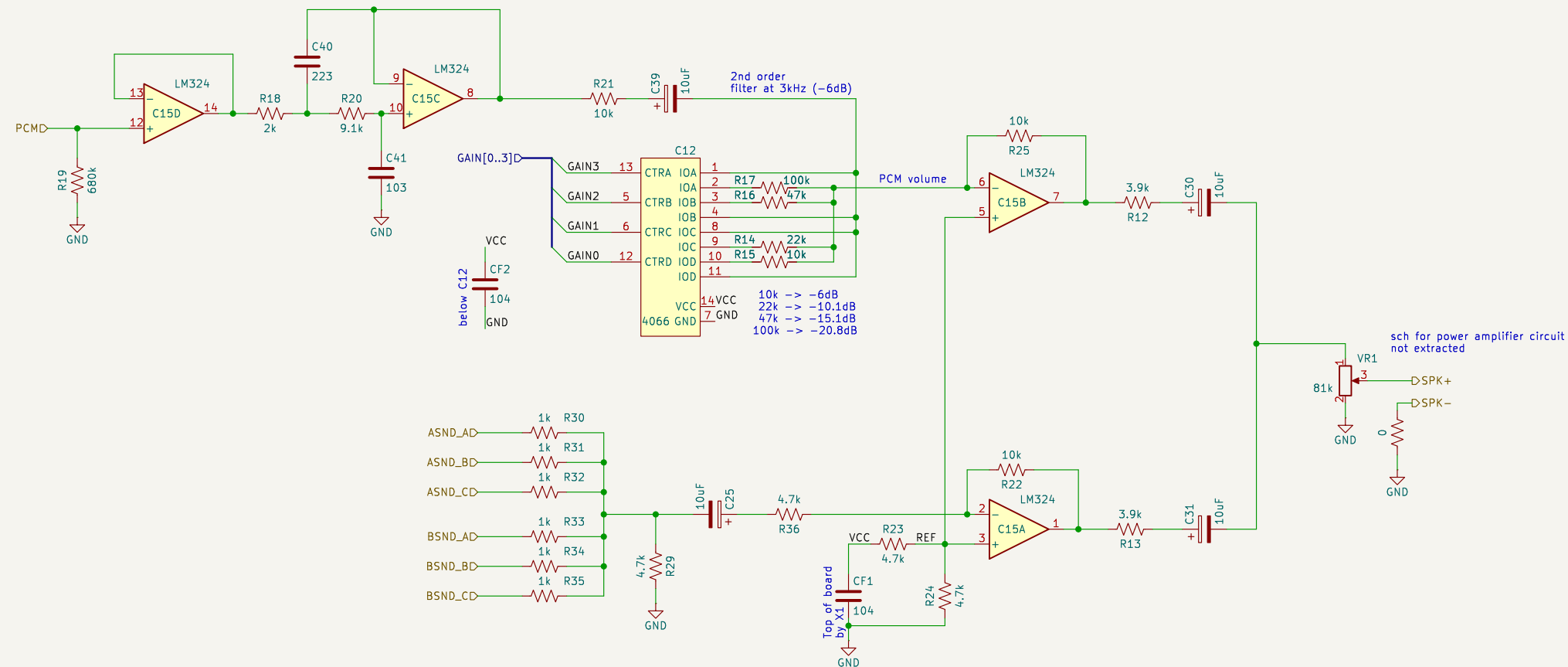
Rev:
Id: 22/26





Title: Watchdog Circuit		
Size: A3	Date: 2024-11-15	Rev:
KiCad E.D.A. 8.0.9		Id: 24/26



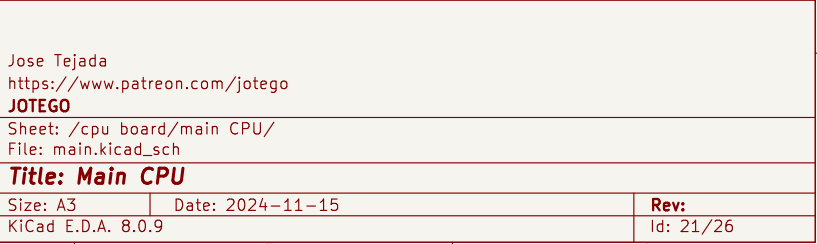


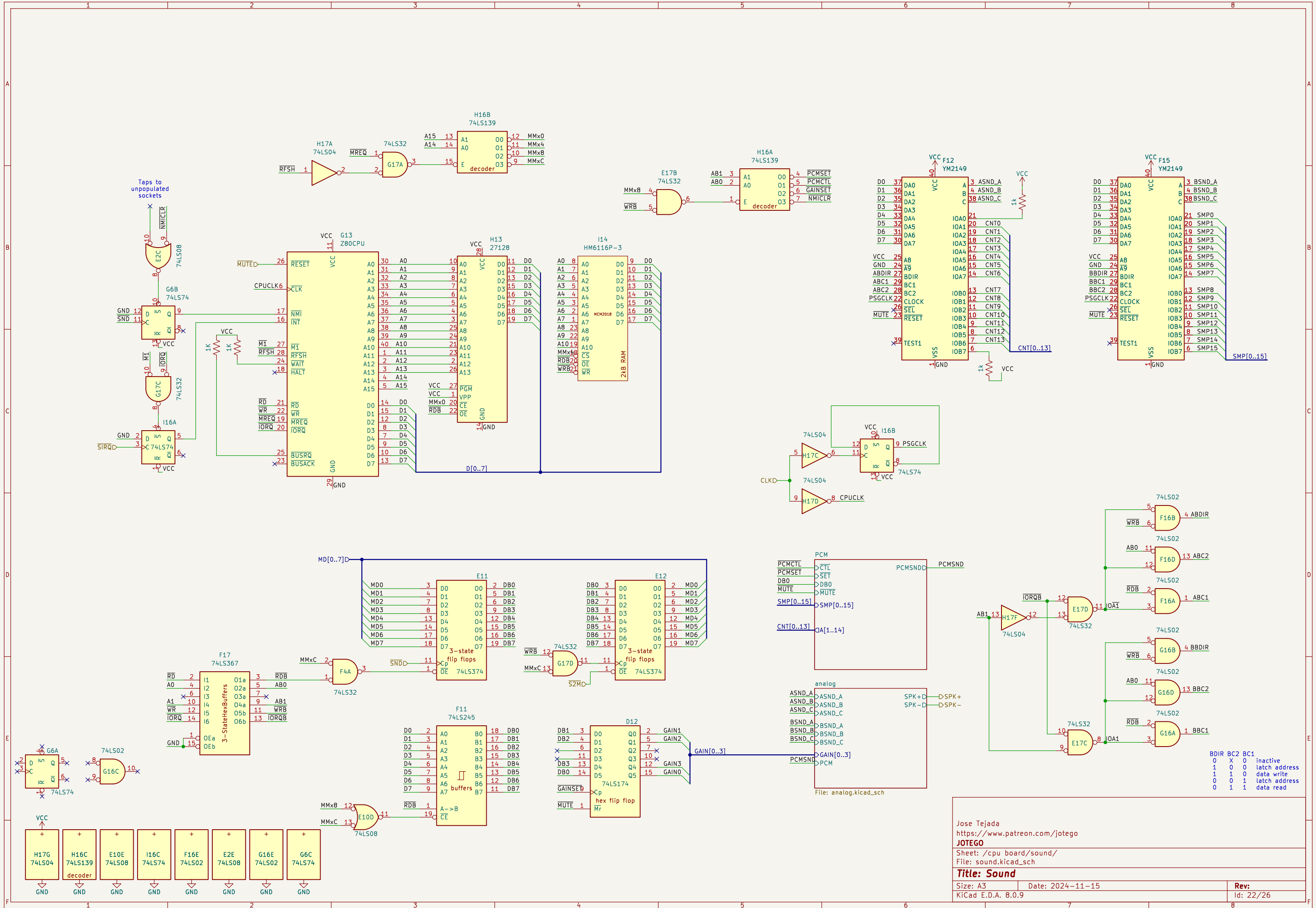
VCC
4
+
C15E
-
V-
11
LM324
GND

Passive component references when shown, come from original PCB. The bootleg did not have references for passives.

Device components follow bootleg references. The bootleg counts letters from left to right and numbers from bottom to top. The original counts numbers left to right and letters bottom to top.

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Sheet: /cpu board/sound/analog/ File: analog.kicad_sch		
Title: Sound amplifiers		
Size: A3	Date: 2024-11-15	Rev:
KiCad E.D.A. 8.0.9		Id: 26/26







Size: A3	Date: 2024-11-15
KiCad E.D.A. 8.0.9	

Size: A3	Date: 2024-11-15	Rev:
KiCad E.D.A. 8.0.9		Id: 23/26

